Probability

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↳ The important terms to be known in probability are.

(i) Random experiment:- A random experiment is one in which all the possible results are known in advance but none of them can be predicted with certainity.

(ii) Outcome:- The results of a random experiment is called an "outcome".

(iii) Sample Space:- The set of all the possible outcomes of a random experiment is called Sample Space & it is denoted by 'S'.

(iv) Event:- A subset of the Sample Space. Consider a simple example. let us say that we toss a coin up in the air. what can happen when it gets back? it will either give a head or a tail. These two are known as outcomes. & the occurance of an outcome is an event. Thus the event is the outcome of some phenomenon.

Probability:- Now let us see the formal definition of probability of an event. let us say for some event E, 'N' is the total number of possible outcomes. For example, if 'E' is a coin toss, then N = 2 i.e., H & T. Out of these possible outcomes, let us say we want to find the probability of some event x (x = occurance of head), that can happen in n (n = 1 = H) ways.

Then we can write the probability of occurance of the event x as.

$$P(X) = \frac{\text{(No. of ways for which 'X' can happen)}}{\text{(Total No. of ways for which the event can Happen)}}$$

$P(x)$ = Probability of occurance of x (head)

$$\therefore P(x) = \frac{n = \text{No. of favourable outcomes}}{N = \text{No. of total possible outcomes.}}$$

↳ Total probability is always = 1 = probabilities of all possible occurances of a random experiments.

↳ Probability of not occurance of X = Total − Probability of occurance of X

$$\Rightarrow \boxed{P(\bar{X}) = 1 - P(X)}$$

Basic Probability Rules:- The probability of an event 'A' is always number b/w 0 & 1, inclusive, i.e.,

$$0 \leq P(A) \leq 1$$

↳ Sometimes probability values are written in %'s

$$\therefore 0\% \leq P(A) \leq 100\%$$

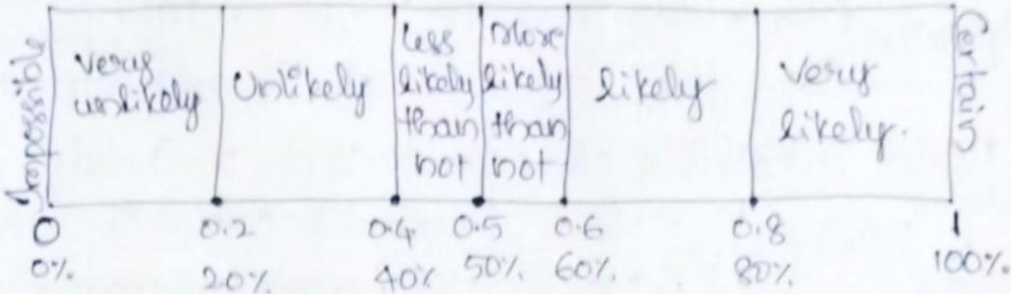
↳ If the event 'A' is not possible, then $P(A) = 0$ (or) $P(A) = 0\%$. If event A is certain to occur, then $P(A) = 1$ (or) $P(A) = 100\%$.

↳ The sum of the probabilities for each possible outcome of an experiment is 1 (or) 100%. Mathematically

$$\sum_{\text{all possible outcomes of A}} P(A) = 1 \quad (\text{or}) \quad \sum_{\text{all possible outcomes of A}} P(A) = 100\%$$

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↳ Probability scale based on the probability value of an occurance of an event ($0 \leq 1$)



↳ To solve Probability mainly we must know about the all possible outcomes of coins, dice & cards based on required random experiment to get required event.

(i) Coins:- when 1 coin tossed total outcomes $= 2^1 = (H, T)$

when 2 coins tossed total outcomes $= \{(H,H), (H,T), (T,H), (T,T)\} = 4 = 2^2$

when n coins tossed total outcomes $= 2^n$

(ii) Dice:- when 1 dice, which is having 6 digits (1, 2, 3, 4, 5, 6) on its 6 faces thrown, total possible outcomes are $= 6 = 6^1 = (1/2/3/4/5/6)$.

when 2 dice thrown total outcomes $= 6^2 = 36$

when n dice thrown total outcomes $= 6^n$

(iii) Cards:- A standard deck of cards has four suits:- hearts, clubs, spades & diamonds. Each suite has 13 cards:- Ace, 2, 3, 4, 5, 6, 7, 8, 9, 10, Jack, Queen, & King. Thus the entire deck / Pack has 52 cards total.

↳ There is no priority of Joker cards (not countable).